Full LC Oscillator Report

1. Optimization Simulation Setup
Item	Value
VDD (V)	1.8
Target Frequency Fo (Hz)	5 GHz
Frequency Offset ΔF (Hz)	1 MHz
Temperature (°C)	27
Process Corner	NA
Inductor Width (μm)	6
Inductor Spacing (μm)	3
Population Size	50
Max Generations	30
Fscale	0.7
Crossover Rate (CR)	0.8
Stall Generation Limit	10
Acceptance Tolerance	-1e9
Stop Cost Threshold	acceptTol
PVT Corners	tt, ss, ff, sf, fs
VDD Sweep (V)	1.62, 1.8, 1.98
Temperature Sweep (°C)	-40, 0, 27, 70, 85

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2. Best Solution Summary
Parameter	Value
Generation	21
Index	0
Total Index	1051
Fosc	5.0806 GHz
Power	9.336 mW
Vpk	1.394 V
Vrms	0.697 V
PN @100k	-95.65 dBc/Hz
PN @1M	-119.51 dBc/Hz
FOM @100k	-180.07
FOM @1M	-183.92

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3. Constraint Verification
Metric	Target	Result	Status
Fosc	4.9–5.1 GHz	5.08 GHz	PASS
Power	≤10 mW	9.34 mW	PASS
Vpk	0.5–1.5 V	1.39 V	PASS
PN @1M	≤ -100 dBc/Hz	-119.5	PASS

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4. PVT Summary (Extremes)
- Frequency: **4.99 – 5.19 GHz**
- Power: **4.17 – 19.2 mW**
- Vpk: **0.84 – 1.89 V**
- PN @1M: **-123.4 to -116 dBc/Hz**
- FOM @1M: **-187.9 to -180.5**

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5. Monte Carlo Summary
Metric	Mean	Sigma
Fosc	5.081 GHz	17.64 MHz
Power	9.36 mW	0.62 mW
PN @1M	-119.5	0.90
Vpk	1.392 V	0.033 V
FOM_1M	-183.9	0.8652

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6. Yield Report
- Total Samples: 1000
- Passing Samples: 704
- **Yield: 70.40%**

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7. Transient (X-Ray) Results
- Oscillation Frequency: **5.072 GHz**
- Differential Swing: **2.74 Vpp**
- RMS Differential: **0.96 V**

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8. Runtime Summary
- Total Runtime: **3.69 hours**